

1. Write the sample space for rolling two dice.

$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6),$
 $(2,1), (2,2), (2,3), (2,4), (2,5), (2,6),$
 $(3,1), (3,2), (3,3), (3,4), (3,5), (3,6),$
 $(4,1), (4,2), (4,3), (4,4), (4,5), (4,6),$
 $(5,1), (5,2), (5,3), (5,4), (5,5), (5,6),$
 $(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$

Sample space = 36.

2. Write the sample space for selecting a traffic light color.

$S = \{\text{Red, Yellow, Green}\}$
Sample space = 3.

3. Identify whether each sample space is finite or Infinite:

- 1) Daily temperature: Infinite
- 2) Deck of cards: Finite
- 3) Student grades (A, B, C, D, F): Finite
- 4) Time until a light bulb fails: Infinite

Sample space = 6

$$\therefore T = 8$$

Sample Space = 3

$$\therefore f = 3$$

$$P = \frac{f}{T} = \frac{3}{6} = \frac{1}{2} = 0.5$$

$\therefore \text{Probability} = 0.5$

5. A card is drawn from a deck. Find the probability of drawing a face card.

$$\therefore \text{Total} = 52$$

Face cards = {Jack, Queen, King}

$\therefore$ Total face cards = $3 \times 4 = 12$

$$\therefore F = 12$$

$$P = \frac{12}{52} = \frac{3}{13} = 0.23$$

$\therefore \text{Probability} = 0.23$

6. A bag contains 5 red, 3 blue, and 2 green balls. Find the Probability of drawing a red ball.

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$$\text{Total balls} = 5 + 3 + 2 = 10$$

$$\therefore \text{Red balls} = 5$$

$$\therefore f = 5$$

$$P = \frac{5}{10} = \frac{1}{2} = 0.5$$

$$\therefore \text{Probability} = 0.5$$

7. IF the Probability of rain is 0.35, what is the Probability that it does not rain?

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$$\text{Formula: } P(A') = 1 - P(A)$$

$$A' = \text{Not rain}$$

$$A = \text{Rain}$$

$$P(\text{Not Rain}) = 1 - P(\text{rain})$$

$$P(\text{Not Rain}) = 1 - 0.35 = 0.65$$

8. Identify whether the following are simple or Compound events:

1) Drawing an Ace from a deck :- Simple Event

2) Rolling an even number on a die :- Compound Event

- 3) Tossing a coin and getting Head :- Simple Event
- 4) Drawing a red or black card :- Compound Event

$$P(\text{Head}) = \frac{1}{2}$$

$$P(\text{Tail}) = \frac{1}{2}$$

$$P(\text{Head or Tail}) = \frac{1}{2} + \frac{1}{2} = 1$$

3) Tossing a coin and getting Head :- Simple Event
4) Drawing a red or black card :- Compound Event

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A) = \frac{1}{2}$$

$$A = \text{Head}$$

$$P(B) = \frac{1}{2}$$

$$B = \text{Tail}$$

$$P(A \cup B) = \frac{1}{2} + \frac{1}{2} - 0 = 1$$

$$C = \text{Red}$$

$$D = \text{Black}$$

$$P(C \cup D) = \frac{1}{2} + \frac{1}{2} - 0 = 1$$

$$P(C \cup D) = P(C) + P(D) - P(C \cap D)$$

$$P(C) = \frac{1}{2} = 0.5, P(D) = \frac{1}{2} = 0.5, P(C \cap D) = 0$$